3 A small object of mass 24 g rests on a platform. The platform is attached to an oscillator, as shown in Fig. 3.1.

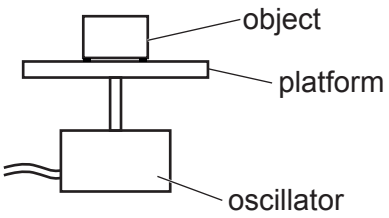


Fig. 3.1

The oscillator moves the platform up and down.

(a) The total energy of the oscillations of the object is 2.2×10^{-4} J.
In one oscillation the object travels a total distance of 14 mm.

Calculate the angular frequency ω of the oscillations.

ω = ... rad s^{-1} [3]

(b) The frequency of the oscillator is fixed, and the amplitude of the oscillations is gradually increased.

(i) Calculate the maximum amplitude of the oscillations so the object does not lose contact with the platform.

amplitude = ... m [2]

(ii) The amplitude of the oscillations is increased so it is greater than the value in **(b)(i)**.

State and explain the position in an oscillation where the object first loses contact with the platform.

.....

.....

.....

..... [2]

[Total: 7]

Markscheme — 9702/42/F/M/24 · Q3

Object on an oscillating platform — angular frequency & amplitude

(a) [3 marks]

Distance travelled per oscillation = $4x_0 \rightarrow x_0 = 14\times10^{-3}/4 = 3.5\times10^{-3}$ m
$E = \frac{1}{2}m\omega^2x_0^2 \rightarrow \omega^2 = 2E/(mx_0^2) = 2\times2.2\times10^{-4}/(0.024\times(3.5\times10^{-3})^2) = 1.50\times10^3$
ω = **39 rad s^{-1}**

(b)(i) [2 marks]

The object loses contact when the maximum downward acceleration of the platform, ω^2x_0, exceeds g:
$x_0 = g/\omega^2 = 9.81/39^2$ = **6.4×10^{-3} m**

(b)(ii) [2 marks]

At the **top of the oscillation** — there the acceleration of the platform is a maximum downwards ($a = -\omega^2x$, magnitude ω^2x_0). The object's maximum possible downward acceleration is g = 9.81 m s^{-2}, so it loses contact when ω^2x_0 exceeds 9.81 m s^{-2}, i.e. first at the top of the oscillation.

4 **(a)** State what is meant by resonance.

.....

.....

..... [2]

(b) A small ball is held in place using a stretched string. One end of the string is fixed to a wall and the other end is attached to a vibration generator, as shown in Fig. 4.1.

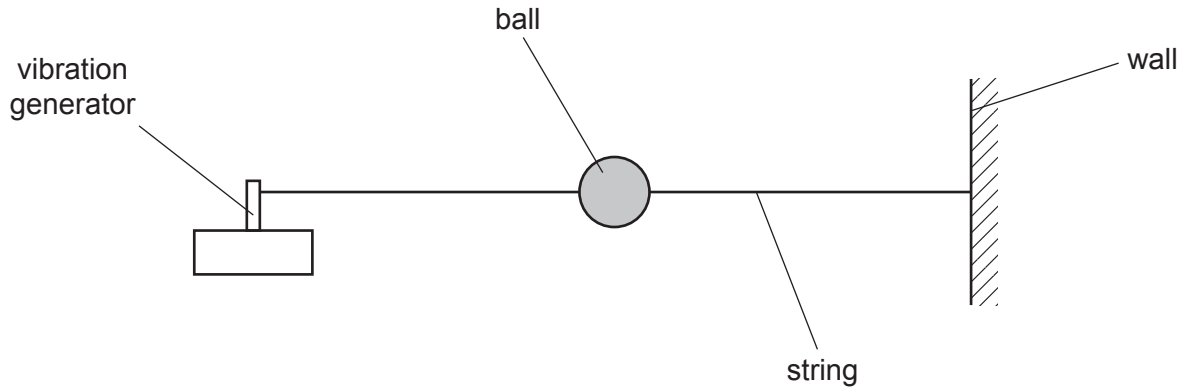


Fig. 4.1

Initially, the vibration generator is switched off.

A student displaces the ball vertically and then releases it. Fig. 4.2 shows the variation of the displacement of the ball with time after it is released.

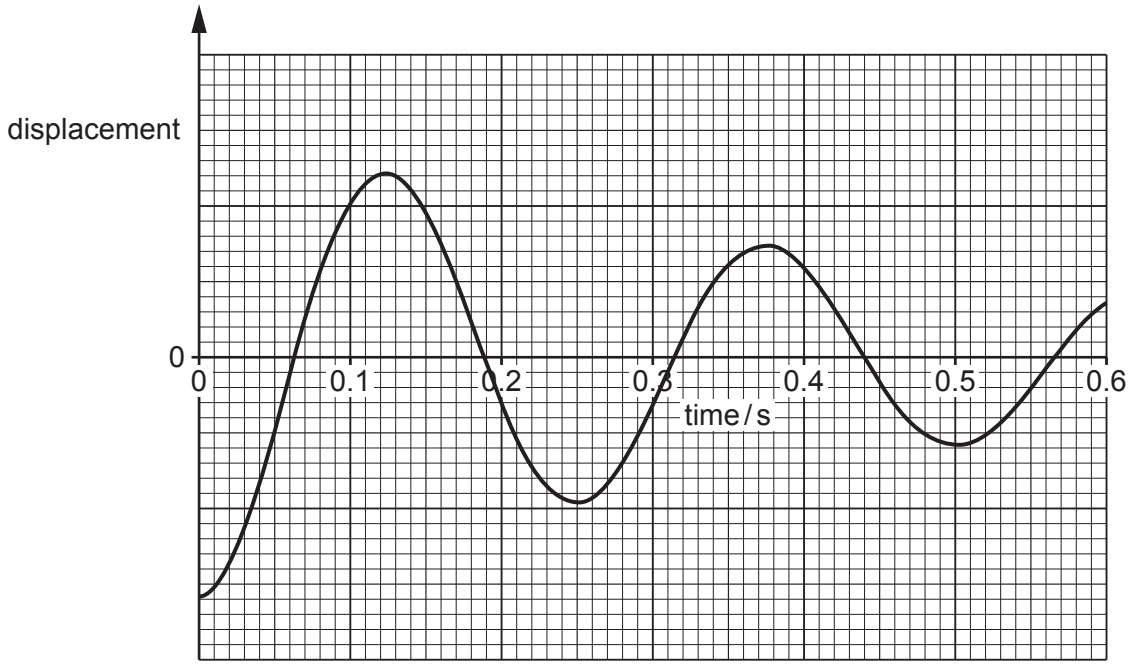


Fig. 4.2

(i) State the name of the phenomenon illustrated by the decrease in the amplitude of the oscillations in Fig. 4.2.

..... [1]

(ii) Explain the decrease with time of the amplitude of the oscillations of the ball.

.....

.....

..... [2]

(iii) Determine the frequency of the oscillations of the ball.

frequency = ... Hz [1]

(c) The vibration generator in **(b)** is switched on and its frequency *f* of vibration is gradually increased from 0 to 10 Hz.

On Fig. 4.3, sketch the variation with *f* of the amplitude of the oscillations of the ball.

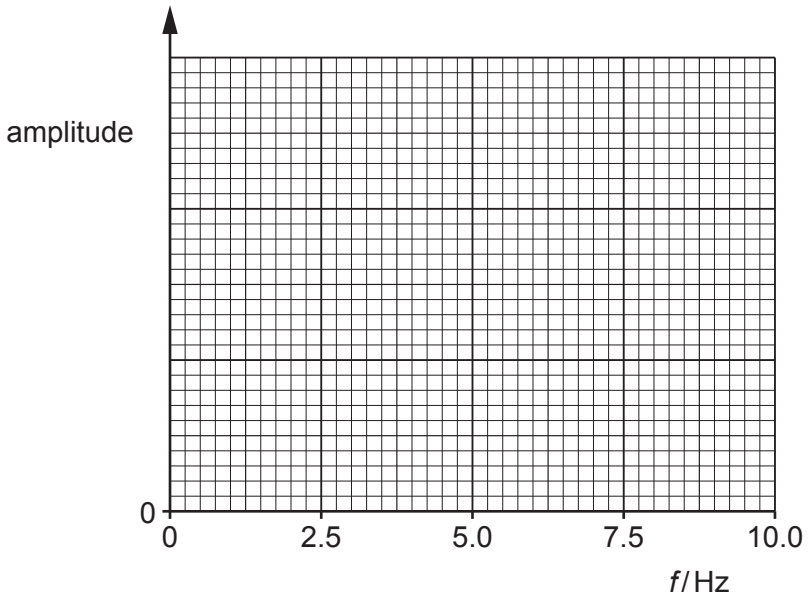


Fig. 4.3

[2]

[Total: 8]

Markscheme — 9702/41/M/J/24 · Q4

Resonance — damping & forced oscillations of a ball

(a) [2 marks]

Resonance is the phenomenon in which an oscillating system has its amplitude a maximum, and this occurs when the driving frequency equals the natural frequency of the system.

(b)(i) [1 mark]

Light damping.

(b)(ii) [2 marks]

The oscillations of the ball lose energy because resistive forces (air resistance / friction in the string) act on the ball and do work against its motion, so the amplitude decreases gradually with time.

(b)(iii) [1 mark]

Period T = 0.25 s (time between successive maxima of Fig. 4.2).
f = 1/T = 1/0.25 = **4.0 Hz**

(c) [2 marks]

A resonance curve: the amplitude rises from zero at f = 0, reaches a single maximum at f = 4.0 Hz (the natural frequency of the ball), and then falls again as f increases towards 10 Hz.

4 A block of mass *m* oscillates vertically on a spring, as shown in Fig. 4.1.

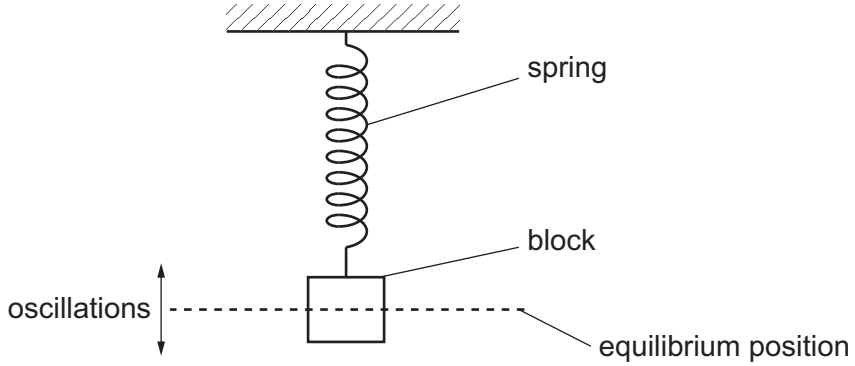


Fig. 4.1

The acceleration *a* of the block varies with displacement *x* from its equilibrium position, as shown in Fig. 4.2.

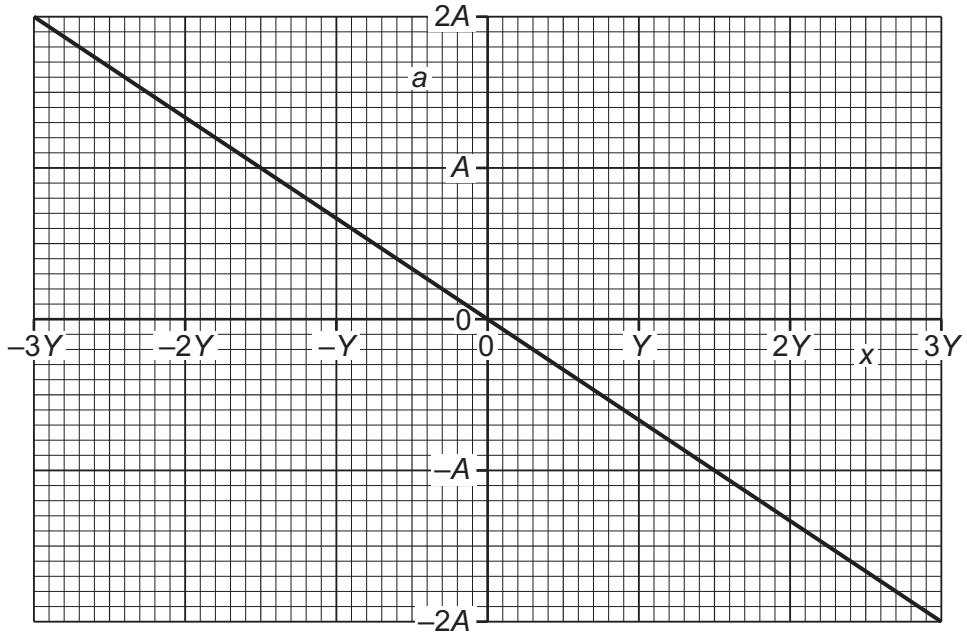


Fig. 4.2

The amplitude of the oscillations is 3*Y* and the maximum acceleration is 2*A*.

(a) Explain how Fig. 4.2 shows that the oscillations of the block are simple harmonic.

.....

.....

..... [2]

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(b) Deduce expressions, in terms of some or all of m, A and Y, for:

(i) the angular frequency ω of the oscillations

ω = ... [1]

(ii) the maximum speed v_0 of the oscillations

v_0 = ... [2]

(iii) the energy E of the oscillations.

E = ... [2]

(c) The period of the oscillations is 0.75 s and the value of $3Y$ is 1.8 cm.

Determine an expression for x in terms of time t, where x is in cm and t is in seconds.

x = ... [2]

[Total: 9]

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Markscheme — 9702/42/M/J/24 · Q4

Block oscillating on a spring — simple harmonic motion & energy

(a) [2 marks]

Fig. 4.2 is a straight line through the origin, showing that **a is proportional to x**; its negative gradient shows that **a and x are always in opposite directions**, i.e. $a = -\omega^2 x$.

(b)(i) [1 mark]

$a = -\omega^2 x$ with $x_0 = 3Y$ and $a_{max} = 2A$:
$2A = \omega^2(3Y)$
$\omega = \mathbf{\sqrt{(2A/3Y)}}$

(b)(ii) [2 marks]

$v_0 = \omega x_0 = \sqrt{(2A/3Y)} \times 3Y$
$= \mathbf{\sqrt{(6AY)}}$

(b)(iii) [2 marks]

$E = \frac{1}{2}m\omega^2 x_0^2 = \frac{1}{2}m \times (2A/3Y) \times (3Y)^2$
$= \frac{1}{2}m \times 6AY = \mathbf{3mAY}$

(c) [2 marks]

$\omega = 2\pi/T = 2\pi/0.75 = 8.4$ rad s^{-1}; amplitude $x_0 = 3Y = 1.8$ cm
$x = x_0 \sin \omega t$
$x = \mathbf{1.8 \sin (8.4t)}$ (x in cm, t in s)

4 **(a)** State what is meant by simple harmonic motion.

.....

.....

..... [2]

(b) A block is suspended from a spring, as shown in Fig. 4.1.

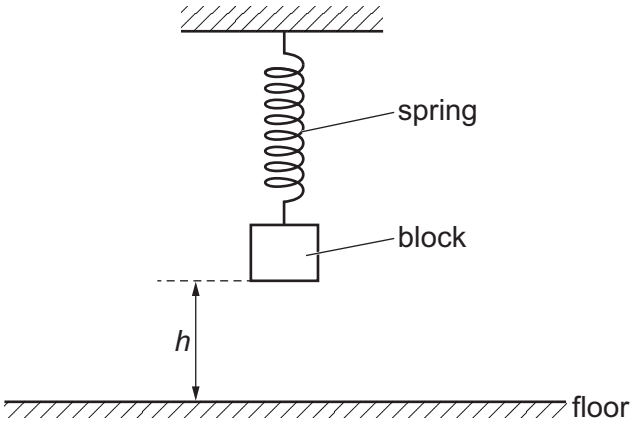


Fig. 4.1

The block is pulled down and released at time $t = 0$. It then oscillates vertically with simple harmonic motion.

Fig. 4.2 shows the variation of the velocity v of the block with height h of the base of the block above the floor.

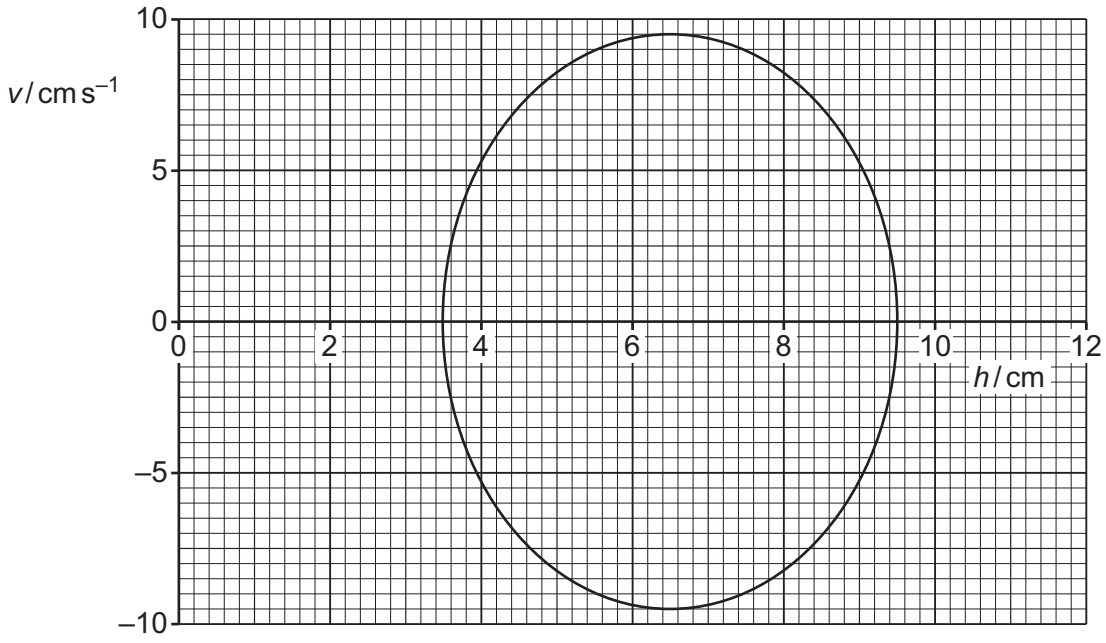


Fig. 4.2

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(i) Determine the amplitude, in cm, of the oscillations.

amplitude = ... cm [1]

(ii) Show that the angular frequency of the oscillations is $3.2\,\text{rad}\,\text{s}^{-1}$.

[2]

(iii) Calculate the period T of the oscillations.

T = ... s [2]

(iv) On Fig. 4.3, sketch the variation of h with time t from $t = 0$ to $t = 6.0\,\text{s}$.

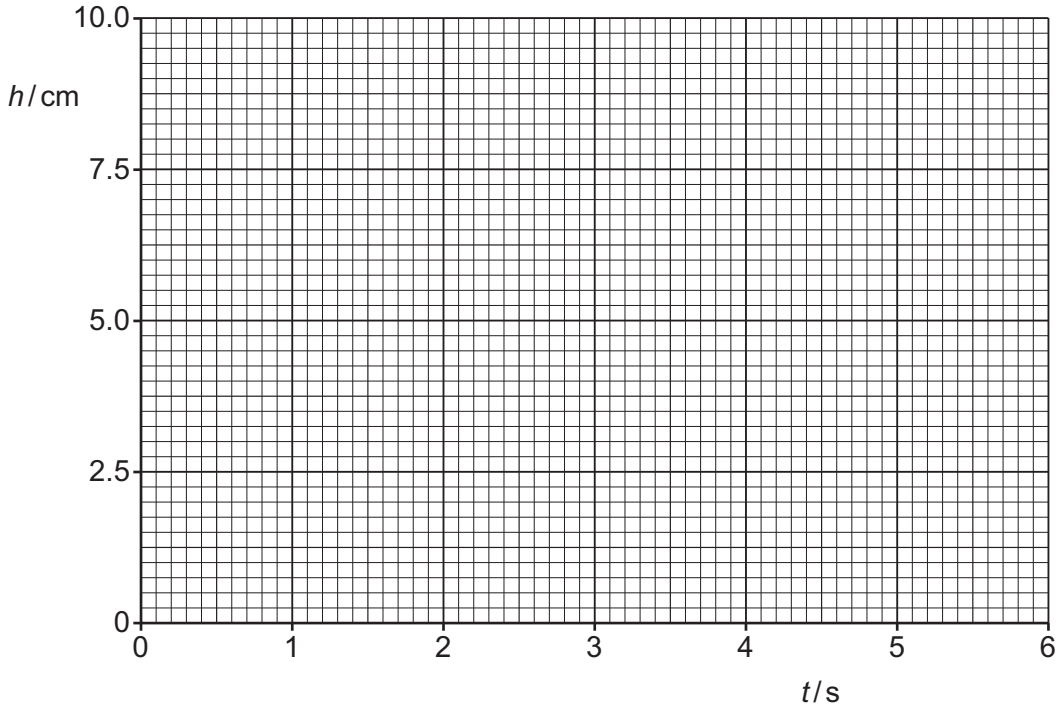


Fig. 4.3

[4]

[Total: 11]

Markscheme — 9702/41/O/N/24 · Q4

Block oscillating on a spring — amplitude, angular frequency & sketch

(a) [2 marks]

Motion in which the acceleration is **directly proportional to the displacement** from a fixed point (the equilibrium position) and is always directed towards that fixed point (opposite in direction to the displacement).

(b)(i) [1 mark]

From Fig. 4.2, the turning points ($v = 0$) are at $h = 3.5$ cm and $h = 9.5$ cm.
amplitude $= (9.5 - 3.5)/2 =$ **3.0 cm**

(b)(ii) [2 marks]

$v_{max} = v_0 = \omega x_0 \rightarrow \omega = v_0/x_0$
$\omega = 9.5/3.0 =$ **3.2 rad s^{-1}**

(b)(iii) [2 marks]

$T = 2\pi/\omega = 2\pi/3.2 =$ **2.0 s**

(b)(iv) [4 marks]

A sinusoidal curve starting at a minimum ($h = 3.5$ cm) at $t = 0$, since the block is released from rest at its lowest point. Three complete cycles from $t = 0$ to $t = 6.0$ s (period 2.0 s), with **all peaks at $h = 9.5$ cm and all troughs at $h = 3.5$ cm**.

5 Fig. 5.1 shows a pendulum consisting of a metal sphere suspended by a thin string.

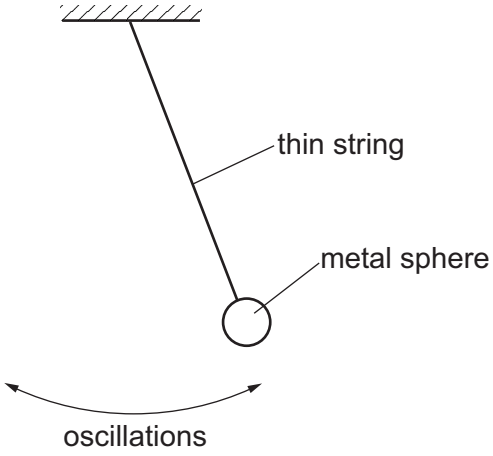


Fig. 5.1 (not to scale)

The sphere undergoes small oscillations about its equilibrium position. The oscillations may be considered to be simple harmonic.

Fig. 5.2 shows the variation with time t of the displacement x of the sphere from its equilibrium position.

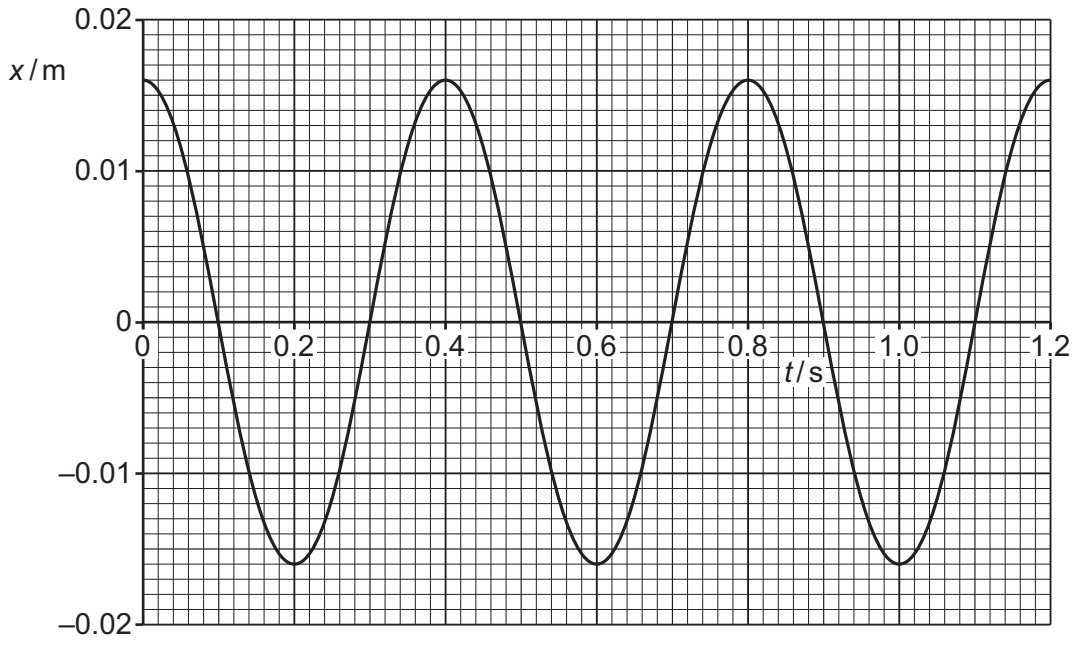


Fig. 5.2

(a) On Fig. 5.1, draw an arrow, from the centre of the sphere, to represent the direction of the resultant force acting on the sphere when it is in the position shown. [1]

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(b) The mass of the sphere is 0.15 kg.

(i) State the amplitude of the oscillations.

amplitude = ... m [1]

(ii) Determine the angular frequency of the oscillations.

angular frequency = ... rad s^{-1} [2]

(iii) Calculate the total energy of the oscillations.

total energy = ... J [2]

(c) On Fig. 5.3, sketch the variation with x of the kinetic energy E_K of the sphere.

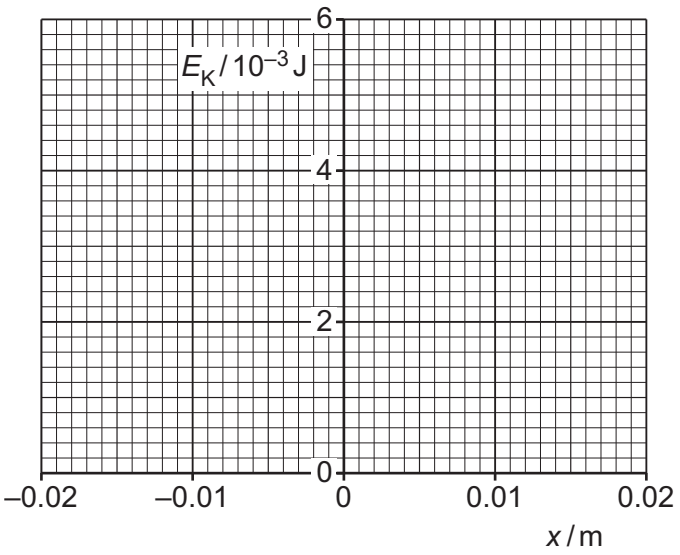


Fig. 5.3

[3]

[Total: 9]

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Markscheme — 9702/42/O/N/24 · Q5

Pendulum in simple harmonic motion — energy & EK sketch

(a) [1 mark]

An arrow drawn from the centre of the sphere, **perpendicular to the string, pointing left and down** — the resultant (restoring) force is directed towards the equilibrium position.

(b)(i) [1 mark]

amplitude = **0.016 m** (maximum displacement, read from Fig. 5.2)

(b)(ii) [2 marks]

T = 0.40 s (from Fig. 5.2)
$\omega = 2\pi/T = 2\pi/0.40$
= **16 rad s^{-1}**

(b)(iii) [2 marks]

$E = \frac{1}{2}m\omega^2 x_0^2 = \frac{1}{2} \times 0.15 \times 15.7^2 \times 0.016^2$
= **4.7×10^{-3} J**

(c) [3 marks]

A dome-shaped curve starting and ending on the x-axis at x = ±0.016 m, rising to a maximum of **4.7×10^{-3} J at x = 0** — kinetic energy is zero at the extremes of the oscillation and maximum at the equilibrium position.

4 A small crystal is made to vibrate with simple harmonic motion. The variation with time t of the displacement x of one surface of the crystal from its equilibrium position is shown in Fig. 4.1.

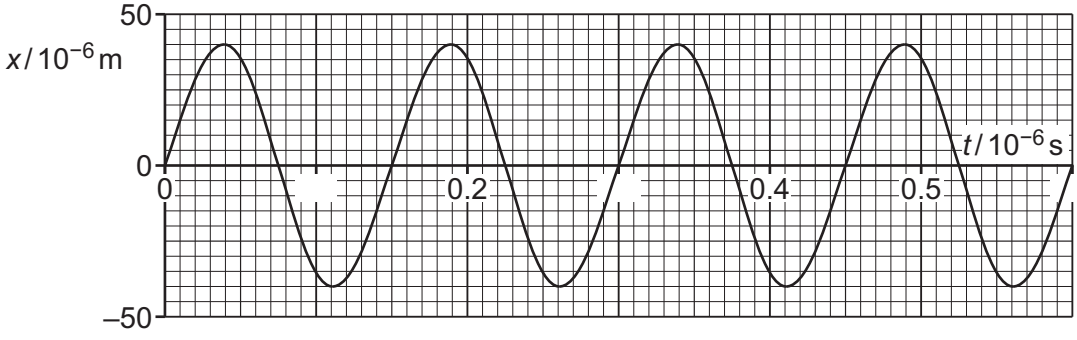


Fig. 4.1

(a) Show that the angular frequency of the vibration of the surface is 4.2×10^{7} rad s^{-1}.

[2]

(b) Determine the maximum acceleration a_0 of the vibration of the surface.

a_0 = ... m s^{-2} [2]

(c) The crystal may be modelled as a single mass of 2.4×10^{-4} kg that vibrates as shown in Fig. 4.1.

Calculate the total energy E of the vibrations.

E = ... J [3]

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(d) The crystal generates ultrasound waves that are used to obtain diagnostic information about internal structures.

(i) The crystal is made from piezoelectric material.

Explain how the crystal is made to vibrate.

.....

.....

.....

..... [2]

(ii) A parallel beam of ultrasound waves is incident on a muscle-bone boundary. Data for muscle and bone are given in Table 4.1.

Table 4.1

material	density / kg m^{-3}	speed of sound / m s^{-1}
muscle	1100	1600
bone	1900	4100

Calculate the percentage of the intensity of the ultrasound beam that is transmitted at this boundary.

percentage transmitted = ... % [3]

[Total: 12]

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Markscheme — 9702/42/F/M/25 · Q4

Piezoelectric crystal — simple harmonic motion & ultrasound

(a) [2 marks]

$\omega = 2\pi/T = 2\pi/(0.15 \times 10^{-6})$
$= \mathbf{4.2 \times 10^{7}\ rad\ s^{-1}}$

(b) [2 marks]

$a_0 = \omega^2 x_0 = (4.2 \times 10^{7})^2 \times 40 \times 10^{-6}$
$= \mathbf{7.1 \times 10^{10}\ m\ s^{-2}}$

(c) [3 marks]

$E = \frac{1}{2} m\omega^2 x_0^2 = \frac{1}{2} \times 2.4 \times 10^{-4} \times (4.2 \times 10^{7})^2 \times (40 \times 10^{-6})^2$
$= \mathbf{340\ J}$

(d)(i) [2 marks]

Apply an **alternating p.d.** across the crystal. The p.d. causes the crystal to **distort** (change shape) at the frequency of the alternating p.d., so the surface vibrates — and conversely an oscillating crystal generates an alternating e.m.f., so it can be used to generate ultrasound.

(d)(ii) [3 marks]

$Z = \rho c$, so $Z_{muscle} = 1100 \times 1600 = 1.76 \times 10^{6}$ and $Z_{bone} = 1900 \times 4100 = 7.79 \times 10^{6}$ kg m^{-2} s^{-1}.
Intensity reflection coefficient $= (Z_1 - Z_2)^2/(Z_1 + Z_2)^2$
$= (7.79 - 1.76)^2/(7.79 + 1.76)^2 = 0.40$ (40%)
so percentage transmitted = **60%**

4 **(a)** State what is meant by simple harmonic motion.

.....

.....

..... [2]

(b) A small sphere is suspended from a fixed point P by a string of negligible mass, as shown in Fig. 4.1.

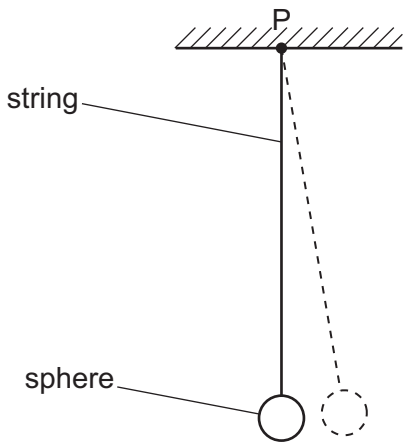


Fig. 4.1

The sphere is given a small horizontal displacement and is then released.

The variation with time of the horizontal velocity v of the sphere is shown in Fig. 4.2.

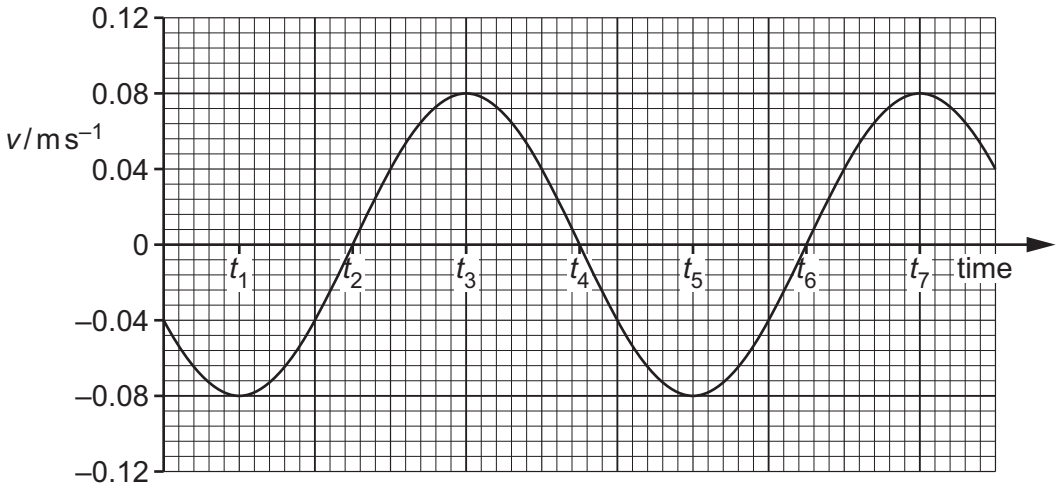


Fig. 4.2

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(i) State two times at which the sphere is passing in the same direction through the equilibrium position.

time ..... and time ..... [1]

(ii) The time interval between t_1 and t_6 is 2.2 s.

Calculate the frequency of oscillation of the sphere.

frequency = ... Hz [2]

(c) The sphere in **(b)** is undergoing simple harmonic motion.

Use your answer in **(b)(ii)** and data from Fig. 4.2 to determine the maximum displacement of the sphere from its equilibrium position.

maximum displacement = ... m [3]

[Total: 8]

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Markscheme — 9702/44/M/J/25 · Q4

Simple pendulum — frequency & maximum displacement

(a) [2 marks]

Motion in which the **acceleration is (directly) proportional to the displacement** and is **always directed towards a fixed point** (in the opposite direction to the displacement).

(b)(i) [1 mark]

The sphere passes through the equilibrium position in the same direction at **t_1 and t_5** (or t_3 and t_7).

(b)(ii) [2 marks]

The interval $t_1 \rightarrow t_6$ is 1.25 periods, so T = 2.2/1.25 = 1.76 s.
f = 1/T = 1.25/2.2 = **0.57 Hz**

(c) [3 marks]

$v_0 = \omega x_0$ with $\omega = 2\pi f$, so $x_0 = v_0/(2\pi f) = 0.080/(2\pi \times 0.57)$
= **0.022 m**

4 **(a)** State what is meant by the frequency of the oscillations of an oscillating object.

.....

..... [1]

(b) An object is oscillating.

Fig. 4.1 shows the variation of the acceleration a of the object with its displacement x from the equilibrium position.

Fig. 4.2 shows the variation of the kinetic energy E_K of the object with time t.

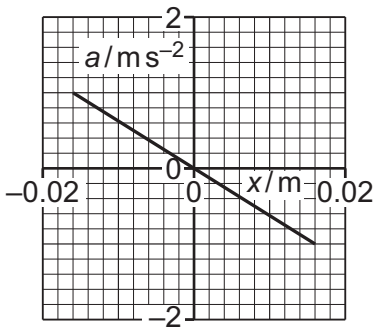


Fig. 4.1

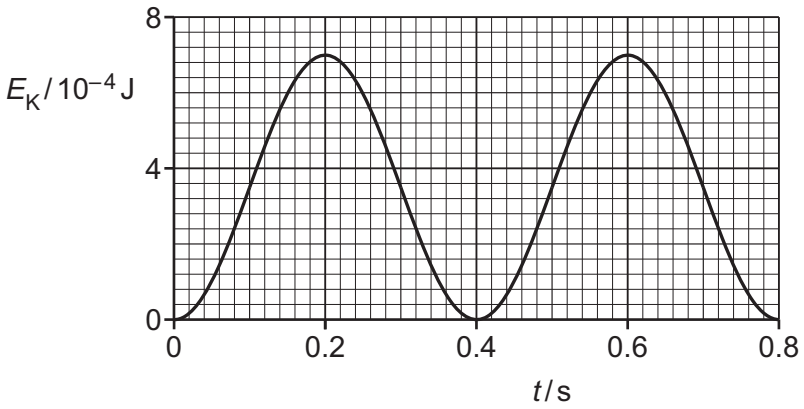


Fig. 4.2

(i) Explain how Fig. 4.2 shows that the period of the oscillations is 0.80 s.

.....

.....

..... [1]

(ii) Calculate the angular frequency ω of the oscillations.

ω = ... rad s^{-1} [2]

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(iii) Apart from the period, frequency and angular frequency of the oscillations, determine **three** other conclusions about the object and its oscillations that may be drawn from Fig. 4.1 and Fig. 4.2. The conclusions may be qualitative or quantitative. Use the space below for any working.

1 .....

.....

2 .....

.....

3 .....

.....

[3]

(iv) Describe the interchange between kinetic energy and potential energy during the oscillations. Numerical values are **not** required.

.....

.....

.....

.....

..... [3]

[Total: 10]

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Markscheme — 9702/44/O/N/25 · Q4

Simple harmonic motion — frequency, energy & interchanges

(a) [1 mark]

The **number of oscillations per unit time**.

(b)(i) [1 mark]

The kinetic energy reaches its maximum/minimum/zero **twice in a cycle**, so one complete cycle takes 0.80 s.

(b)(ii) [2 marks]

$\omega = 2\pi/T = 2\pi/0.80$
$= \mathbf{7.9\ rad\ s^{-1}}$ (or $\omega = \sqrt{(a_0/x_0)} = \sqrt{(1.0/0.016)}$)

(b)(iii) [3 marks]

Any three of: the **oscillations are simple harmonic**; the **amplitude = 0.016 m**; the **maximum speed = 0.13 m s^{-1}**; the **total energy = 7.0 × 10^{-4} J**; the **mass of the object = 0.087 kg**; the **maximum momentum = 0.011 kg m s^{-1}**.

(b)(iv) [3 marks]

The kinetic energy is a **maximum at zero displacement** and **zero at maximum displacement**, while the potential energy is **zero at zero displacement** and a **maximum at maximum displacement**. The **sum of the kinetic and potential energies is constant** (total energy).

5 A cuboidal block floats in a liquid with its base horizontal, as shown in Fig. 5.1.

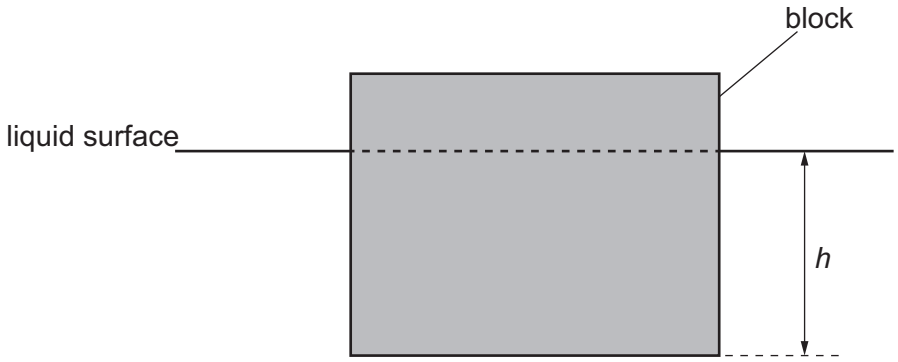


Fig. 5.1

The base of the block is at a depth h below the surface of the liquid.

The block is displaced downwards by a small distance and then released so that it oscillates.

Fig. 5.2 shows the variation with h of the acceleration a of the block.

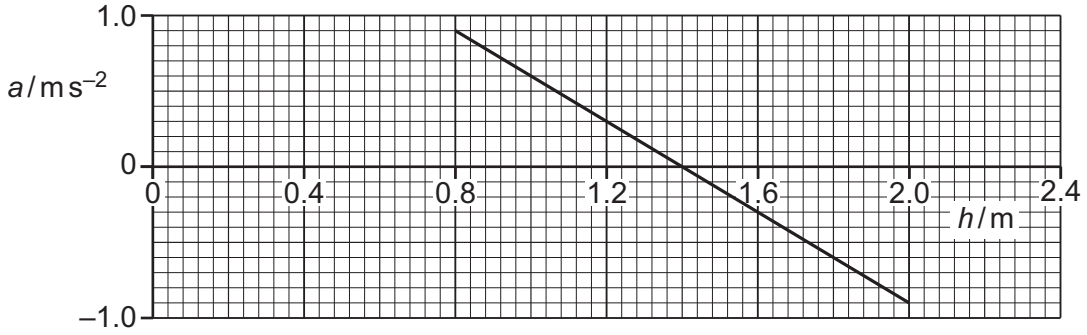


Fig. 5.2

Fig. 5.3 shows the variation with h of the kinetic energy E_K of the block.

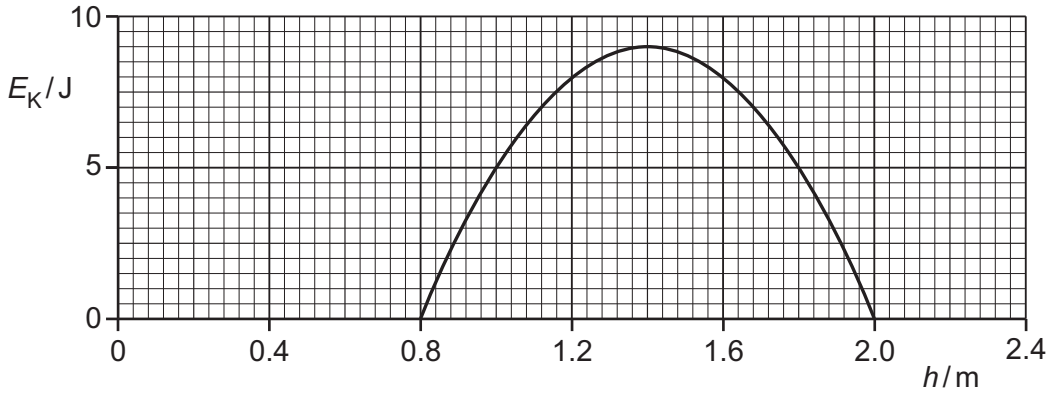


Fig. 5.3

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(a) **(i)** Determine the amplitude of the oscillations.

amplitude = ... m [1]

(ii) State what the line in Fig. 5.2 shows about the nature of the oscillations.

..... [1]

(b) State **three** other quantitative conclusions that can be drawn from Fig. 5.2 and Fig. 5.3 about the block and its oscillations. Use the space for any working.

1 .....

.....

2 .....

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3 .....

.....

[3]

(c) On Fig. 5.4, sketch the variation with h of the potential energy E_P of the oscillations.

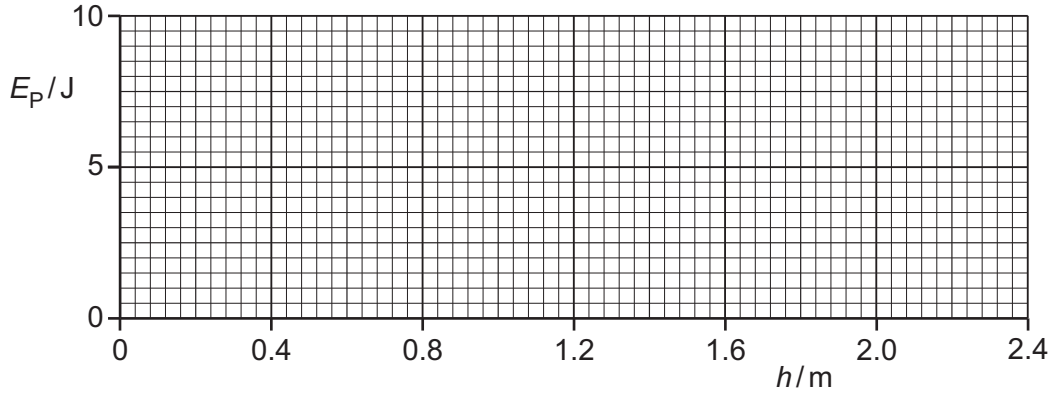


Fig. 5.4

[3]

[Total: 8]

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Markscheme — 9702/41/M/J/25 · Q5

Floating block — simple harmonic motion & energy

(a)(i) [1 mark]

The oscillation extends from h = 0.8 m to h = 2.0 m about the equilibrium at h = 1.4 m, so the amplitude = **0.60 m**.

(a)(ii) [1 mark]

The line is a straight line through the origin, showing $a \propto -$(displacement), so the **oscillations are simple harmonic**.

(b) [3 marks]

Any three of: the **mean/equilibrium position is at h = 1.4 m**; the **total energy of the oscillations is 9.0 J**; the **angular frequency is 1.2 rad s^{-1}** (period 5.1 s, or frequency 0.19 Hz); the **maximum speed is 0.73 m s^{-1}**; the **mass of the block is 33 kg**.

(c) [3 marks]

A **U-shaped curve** resting on the h-axis (minimum at $E_P = 0$), drawn between h = 0.8 m and h = 2.0 m with the **minimum at h = 1.4 m** and both end-points at **E_P = 9.0 J** (since E_P = total energy − E_K).

5 **(a)** State what is meant by simple harmonic motion.

.....

.....

..... [2]

(b) A block is suspended by a spring. The block oscillates vertically with simple harmonic motion.

The velocity v of the block varies with time t according to

$$v = 0.56 \cos 16t$$

where v is in m s^{-1} and t is in s.

(i) Calculate the period of the oscillation.

period = ... s [1]

(ii) Determine the amplitude x_0 of the oscillation.

x_0 = ... m [2]

(iii) Use your answer in **(b)(ii)** to determine the equation for v in terms of the displacement x of the block, where v is in m s^{-1} and x is in m.

v = ... [1]

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(iv) On Fig. 5.1, sketch the variation of v with x.

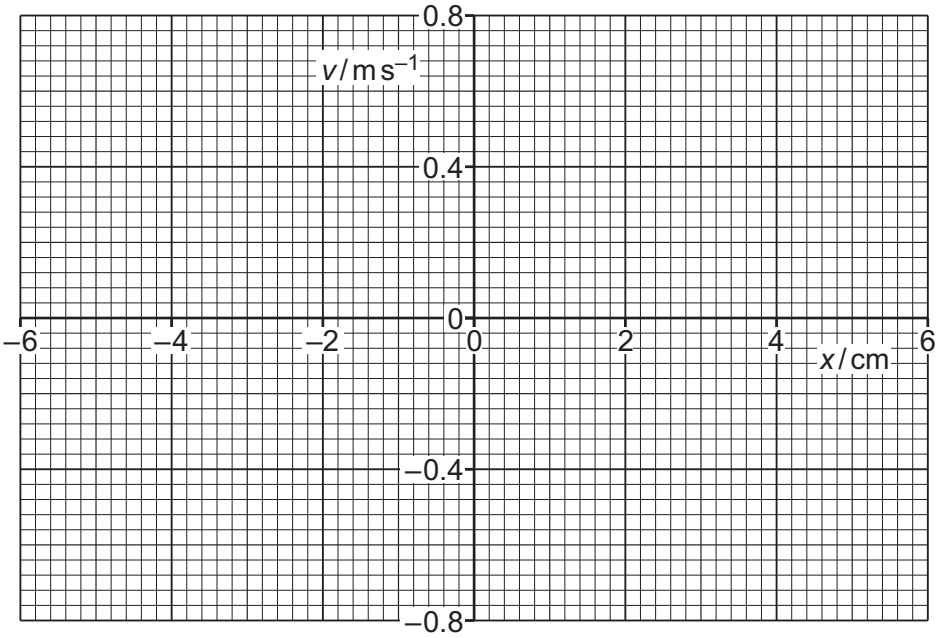


Fig. 5.1

[3]

[Total: 9]

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Markscheme — 9702/42/M/J/25 · Q5

Simple harmonic motion — displacement, velocity & phase

(a) [2 marks]

Motion in which the **acceleration is (directly) proportional to the displacement** and is **always directed towards a fixed point** (in the opposite direction to the displacement).

(b)(i) [1 mark]

period = $2\pi/\omega = 2\pi/16$
= **0.39 s**

(b)(ii) [2 marks]

$v_0 = \omega x_0$, so $x_0 = v_0/\omega = 0.56/16$
= **0.035 m**

(b)(iii) [1 mark]

$v = \pm\omega\sqrt{(x_0^2 - x^2)}$ = **$\pm 16\sqrt{(0.035^2 - x^2)}$**

(b)(iv) [3 marks]

A **closed loop (ellipse) surrounding the origin**, crossing the x-axis (v = 0) at **x = ±3.5 cm** and crossing the v-axis (x = 0) at **v = ±0.56 m s^{-1}**.

5 A cuboidal block floats in a liquid with its base horizontal, as shown in Fig. 5.1.

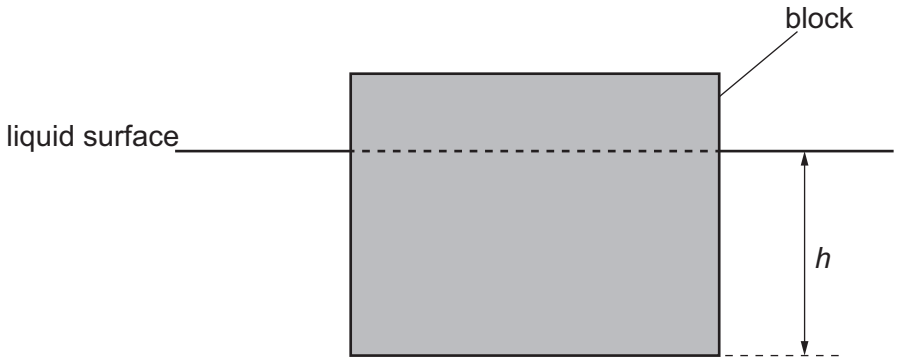


Fig. 5.1

The base of the block is at a depth h below the surface of the liquid.

The block is displaced downwards by a small distance and then released so that it oscillates.

Fig. 5.2 shows the variation with h of the acceleration a of the block.

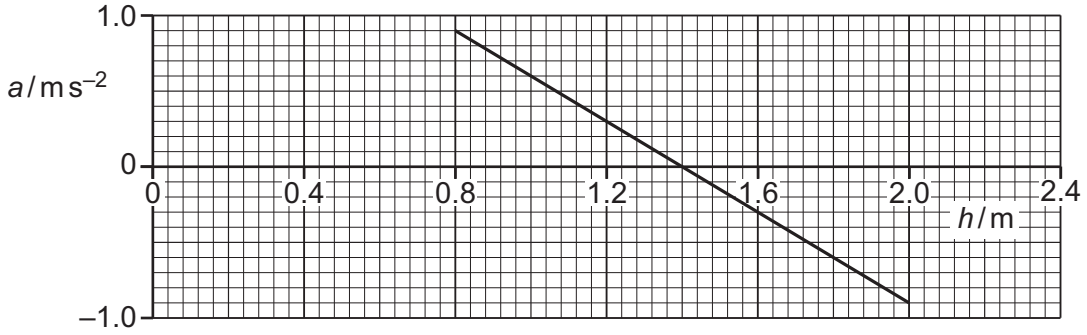


Fig. 5.2

Fig. 5.3 shows the variation with h of the kinetic energy E_K of the block.

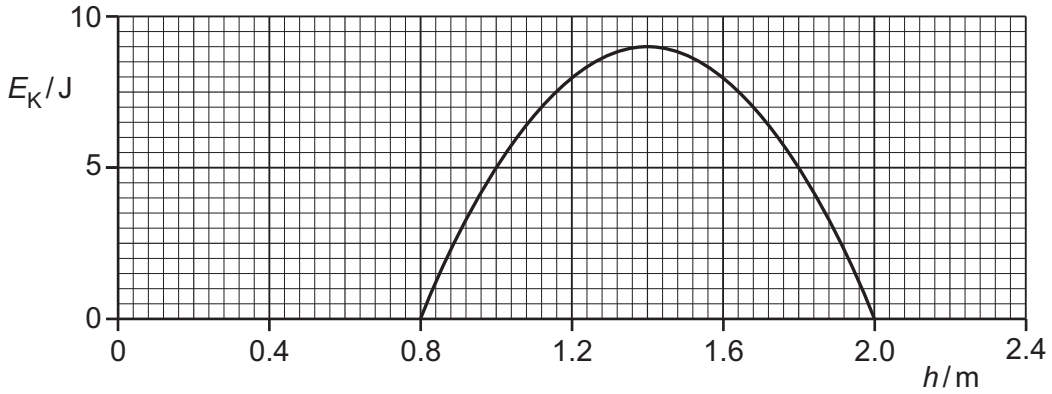


Fig. 5.3

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(a) **(i)** Determine the amplitude of the oscillations.

amplitude = ... m [1]

(ii) State what the line in Fig. 5.2 shows about the nature of the oscillations.

..... [1]

(b) State **three** other quantitative conclusions that can be drawn from Fig. 5.2 and Fig. 5.3 about the block and its oscillations. Use the space for any working.

1 .....

.....

2 .....

.....

3 .....

.....

[3]

(c) On Fig. 5.4, sketch the variation with h of the potential energy E_P of the oscillations.

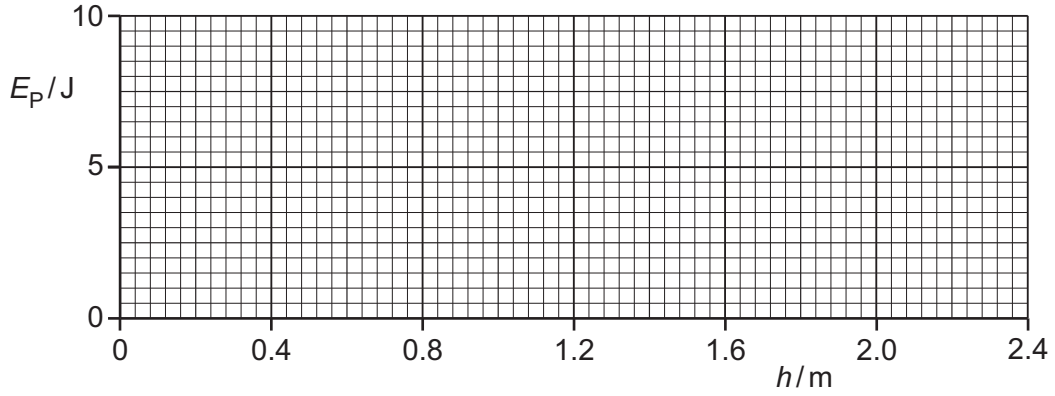


Fig. 5.4

[3]

[Total: 8]

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Markscheme — 9702/43/M/J/25 · Q5

Floating block — simple harmonic motion & energy

(a)(i) [1 mark]

The oscillation extends from h = 0.8 m to h = 2.0 m about the equilibrium at h = 1.4 m, so the amplitude = **0.60 m**.

(a)(ii) [1 mark]

The line is a straight line through the origin, showing a ∝ −(displacement), so the **oscillations are simple harmonic**.

(b) [3 marks]

Any three of: the **mean/equilibrium position is at h = 1.4 m**; the **total energy of the oscillations is 9.0 J**; the **angular frequency is 1.2 rad s⁻¹** (period 5.1 s, or frequency 0.19 Hz); the **maximum speed is 0.73 m s⁻¹**; the **mass of the block is 33 kg**.

(c) [3 marks]

A **U-shaped curve** resting on the h-axis (minimum at $E_P = 0$), drawn between h = 0.8 m and h = 2.0 m with the **minimum at h = 1.4 m** and both end-points at $\mathbf{E_P = 9.0\ J}$ (since E_P = total energy − E_K).

5 A steel ball on the end of a thin string oscillates with small oscillations, as shown in Fig. 5.1.

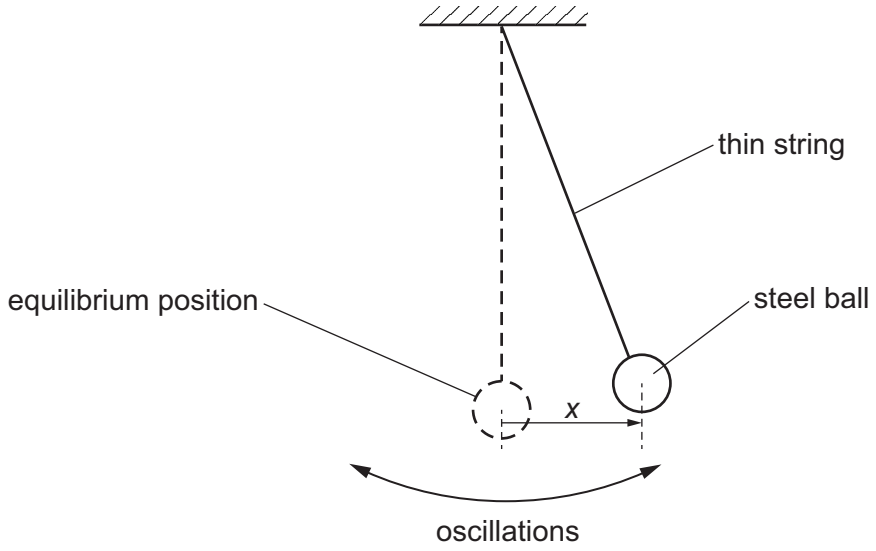


Fig. 5.1 (not to scale)

The displacement of the centre of the ball from its equilibrium position is x.

(a) Fig. 5.2 shows the variation with x of the acceleration a of the ball.

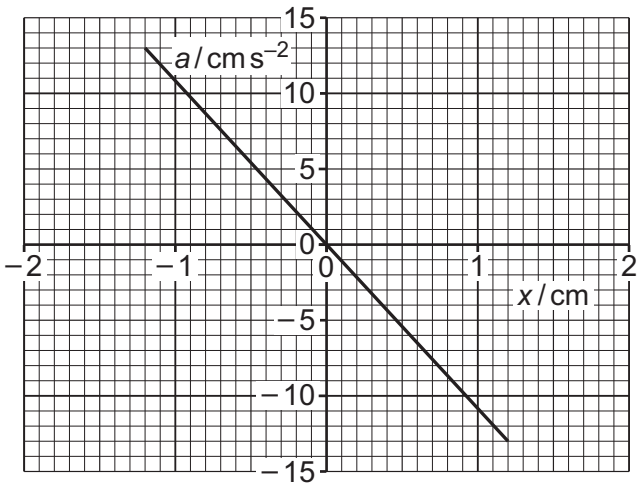


Fig. 5.2

(i) Explain how Fig. 5.2 shows that the oscillations of the ball are simple harmonic.

.....

.....

..... [2]

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(ii) Determine the period T of the oscillations.

T = ... s [3]

(b) At time $t = 0$, when the displacement of the ball has its maximum value, the ball is immersed in a trough containing thick oil so that the ball is just below the surface of the oil. This results in the subsequent motion of the ball being heavily damped.

(i) State what is meant by damping.

.....

.....

..... [2]

(ii) On Fig. 5.3, sketch a possible variation of the displacement x of the ball with t between $t = 0$ and $t = 2T$.

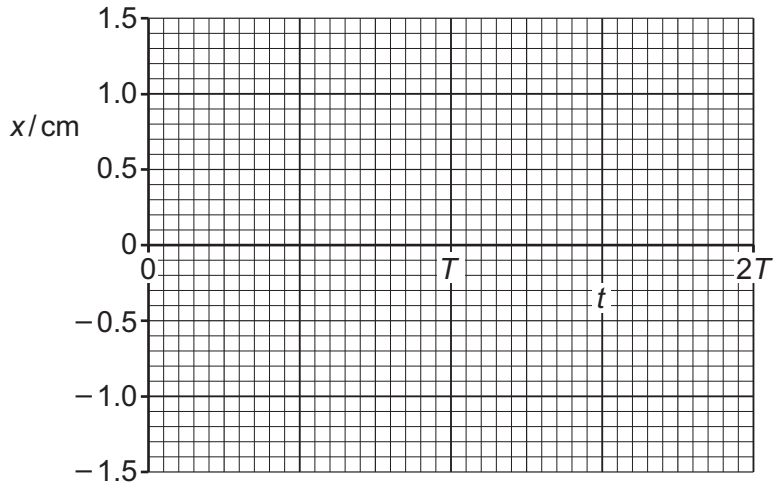


Fig. 5.3

[3]

[Total: 10]

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Markscheme — 9702/42/O/N/25 · Q5

Simple harmonic motion & heavy damping — steel ball in oil

(a)(i) [2 marks]

The graph is a **straight line through the origin**, showing a is proportional to x, and its **negative gradient** shows a is always in the opposite direction to x — so the oscillations are simple harmonic.

(a)(ii) [3 marks]

$a_0 = \omega^2 x_0$ and $\omega = 2\pi/T$, so $T = 2\pi\sqrt{(x_0/a_0)} = 2\pi\sqrt{(1.2/13)}$
= **1.9 s**

(b)(i) [2 marks]

Damping is the **loss of energy from the oscillations** due to **resistive forces**.

(b)(ii) [3 marks]

A line starting at x = ±1.2 cm at t = 0 (entirely above or below the t-axis), with both the magnitude of x and the magnitude of the gradient **continuously decreasing** — heavy damping, so the ball returns to equilibrium without oscillating.

7 A bar magnet is suspended from a spring. One pole of the magnet oscillates freely in a coil of wire, as shown in Fig. 7.1.

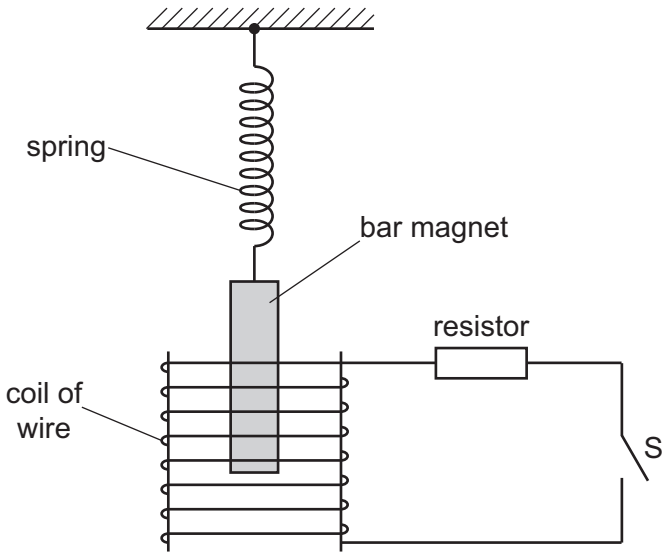


Fig. 7.1

The switch S is initially open.

(a) The switch S is now closed. As a result, the oscillations of the magnet are lightly damped.

(i) State what is meant by damping.

.....

.....

..... [2]

(ii) Describe what is observed to indicate that the damping is light.

.....

..... [1]

(iii) By reference to electromagnetic induction and to conservation of energy, explain why the oscillations are damped.

.....

.....

.....

..... [3]

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(b) The procedure in **(a)** is repeated after replacing the resistor with one of greater resistance.

Suggest, with a reason, the effect of this change on the oscillations.

.....

.....

..... [2]

[Total: 8]

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Markscheme — 9702/44/M/J/25 · Q7

Damped oscillations — magnet in a coil & electromagnetic induction

(a)(i) [2 marks]

Damping is the **loss of energy from the oscillations** due to **resistive forces**.

(a)(ii) [1 mark]

The **amplitude of the oscillations decreases gradually** (the oscillations continue for several periods).

(a)(iii) [3 marks]

The oscillating magnet **cuts magnetic flux**, inducing an e.m.f. in the coil. This **drives a current through the resistor**, which **dissipates thermal energy**. By conservation of energy this thermal energy comes from the energy of the oscillations, so the oscillations are damped.

(b) [2 marks]

For a given induced e.m.f. the **current is smaller**, so thermal energy is dissipated at a lower rate (the resistive force is smaller). The oscillations are therefore **less damped** — the amplitude decreases more slowly and they continue for longer.

5 A block of mass 27 g is suspended from a spring so that it rests in equilibrium, as shown in Figure 5.1.

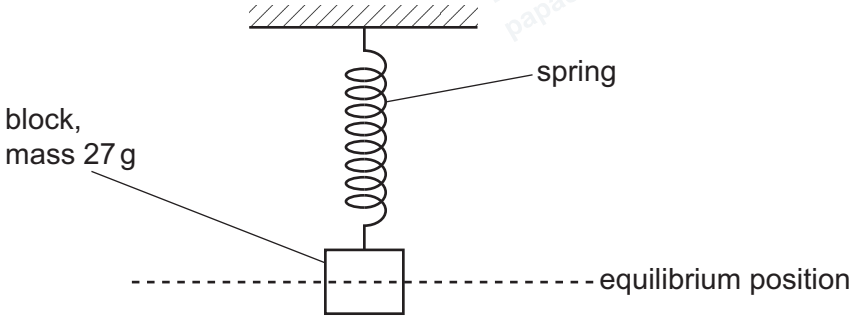


Figure 5.1

The block is displaced downwards through a small distance. The block is released and then oscillates vertically with simple harmonic motion.

The oscillations have a period of 1.7 s and an energy of 13 μJ.

(a) (i) Show that the angular frequency of the oscillations is 3.7 rad s^{-1}.

[1]

(ii) State what is meant by the amplitude of the oscillations.

.....

.....

..... [1]

(iii) Calculate the amplitude of the oscillations.

amplitude = ... m [3]

(b) On Figure 5.2, sketch the variation of the kinetic energy E_K of the block with displacement x from the equilibrium position for one complete oscillation of the block. Add a suitable scale to the x-axis.

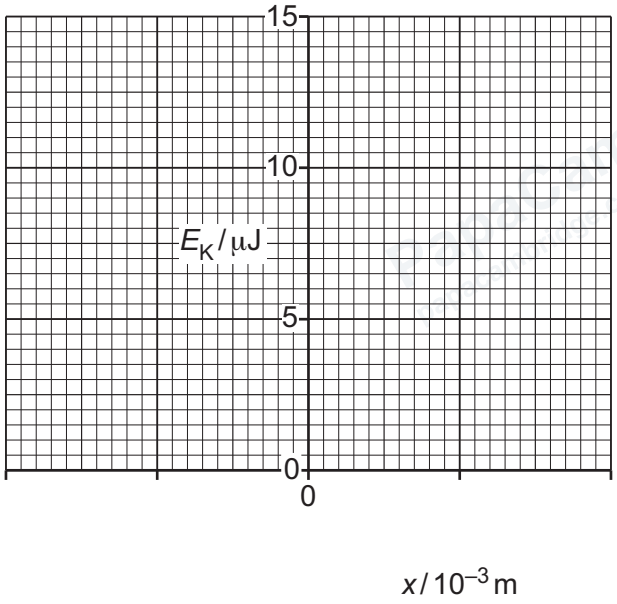


Figure 5.2

[3]

[Total: 8]

Markscheme — 9702/42/M/J/26 · Q5

Block oscillating on a spring — amplitude & kinetic energy

(a)(i) [1 mark]

$\omega = 2\pi/T = 2\pi/1.7$
$= \mathbf{3.7\ rad\ s^{-1}}$

(a)(ii) [1 mark]

The amplitude is the **magnitude of the maximum displacement of the block from its equilibrium (mean) position**.

(a)(iii) [3 marks]

$E = \frac{1}{2}m\omega^2 x_0^2$
$13\times10^{-6} = \frac{1}{2}\times27\times10^{-3}\times3.7^2\times x_0^2$
$x_0^2 = 2\times13\times10^{-6}/(0.027\times3.7^2) = 7.03\times10^{-5}$
$x_0 = \mathbf{8.4\times10^{-3}\ m}$

(b) [3 marks]

Sketch a dome-shaped curve, symmetrical about the EK-axis, with a single peak of **13 μJ at x = 0**, falling to EK = 0 at $x = \pm8.4\times10^{-3}$ m (the amplitude). Add a consistent x-axis scale with at least one marking in each half (e.g. ±8 and ±4, in units of 10^{-3} m), corresponding to the end points.

5 A metal ball is attached to the lower end of a spring, as shown in Figure 5.1.

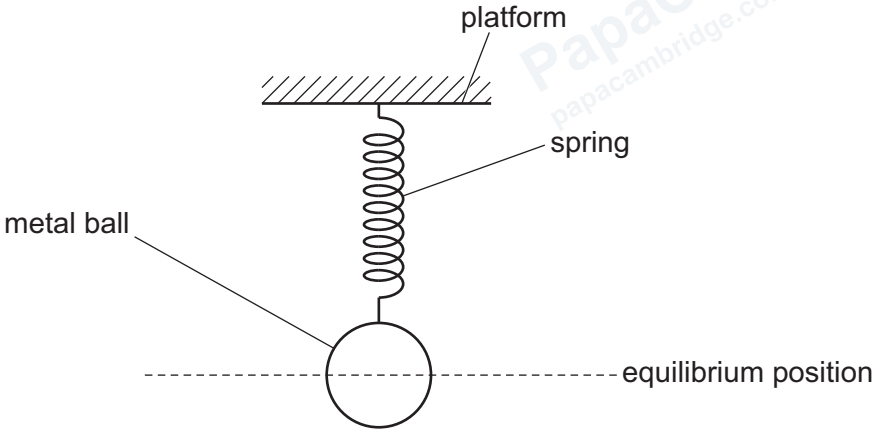


Figure 5.1

The upper end of the spring is attached to a platform that is initially stationary.

(a) The ball is pulled down by a small distance and released so that it oscillates with simple harmonic motion. The oscillations have an amplitude of 2.5 cm and a maximum speed of 4.4 cm s^{-1}.

(i) On Figure 5.2, sketch the variation of the velocity v of the ball with the displacement x from its equilibrium position as the ball undergoes one full oscillation.

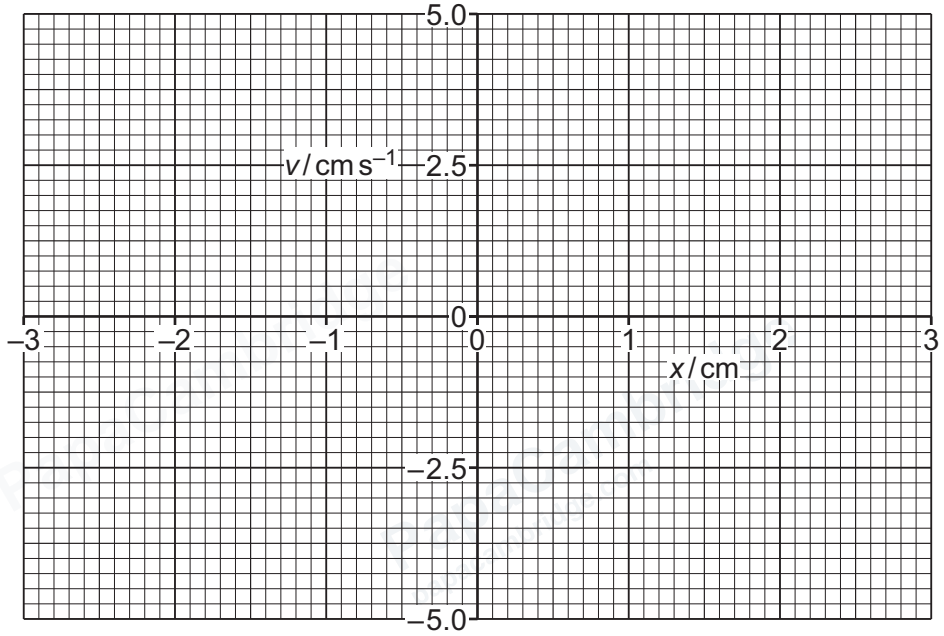


Figure 5.2

[3]

(ii) Show that the frequency of the oscillations is 0.28 Hz.

[2]

(b) The ball is returned to rest at its equilibrium position.

The platform is now moved up and down by a motor so that it oscillates vertically with simple harmonic motion of constant amplitude but variable frequency. The oscillating platform causes the ball to oscillate.

(i) State what is meant by resonance.

.....

.....

..... [2]

(ii) The frequency f of the oscillations of the platform slowly increases from 0 to 0.40 Hz.

On Figure 5.3, sketch the variation with f of the amplitude of the oscillations of the **ball**.

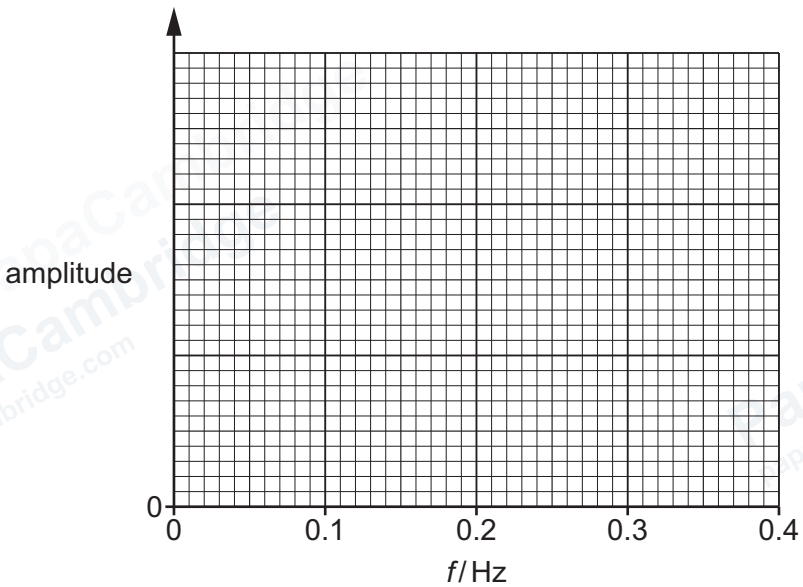


Figure 5.3

[2]

[Total: 9]

Markscheme — 9702/44/M/J/26 · Q5

Ball on a spring — velocity–displacement graph & resonance

(a)(i) [3 marks]

Draw a smooth closed loop (ellipse) enclosing the origin, cutting the x-axis at x = **±2.5 cm** (and nowhere further out) and the v-axis at v = **±4.4 cm s⁻¹** (and nowhere further out). The v–x graph for s.h.m. is an ellipse, $v = \pm\omega\sqrt{(x_0^2 - x^2)}$, not straight lines between the intercepts.

(a)(ii) [2 marks]

$v_0 = \omega x_0$ and $\omega = 2\pi f$:
$f = v_0/(2\pi x_0) = 4.4/(2\pi \times 2.5) = 4.4/15.7$
= **0.28 Hz**

(b)(i) [2 marks]

Resonance is the oscillation of an object at **maximum amplitude**, which occurs when the **driving frequency equals the natural frequency** of the oscillating system.

(b)(ii) [2 marks]

Draw a curve rising continuously from f = 0 (amplitude never zero when f > 0) to a single peak at f = **0.28 Hz**, then falling continuously to f = 0.40 Hz.